

Introduction to Agentic AI & Large Language Model (LLM) Systems

Artificial Intelligence (AI) has transformed the way humans interact with technology. From voice assistants and recommendation systems to autonomous vehicles and intelligent healthcare applications, AI has become an integral part of modern society. Over the past decade, AI has evolved rapidly from traditional rule-based systems and machine learning models to advanced deep learning architectures capable of understanding and generating human-like language. One of the most significant breakthroughs in this evolution is the development of **Large Language Models (LLMs)** and **Agentic AI**.

A **Large Language Model (LLM)** is an advanced artificial intelligence model trained on massive amounts of text data collected from books, research papers, websites, and other digital resources. Using the Transformer architecture, LLMs learn patterns, grammar, reasoning abilities, and contextual relationships within language. As a result, they can perform a wide variety of natural language tasks such as answering questions, generating content, summarizing documents, translating languages, writing computer programs, and assisting users in solving complex problems. Popular examples of LLMs include **ChatGPT (OpenAI)**, **Gemini (Google)**, **Claude (Anthropic)**, and **Llama (Meta)**.

While LLMs are highly capable of understanding and generating text, they primarily operate in a **reactive** manner—they respond to prompts provided by users. They do not independently plan, make decisions, or execute multi-step tasks unless explicitly instructed. To overcome these limitations, the field of **Agentic AI** has emerged.

Agentic AI represents the next generation of artificial intelligence, where intelligent systems behave more like autonomous assistants than simple chatbots. Instead of only generating responses, Agentic AI systems can understand goals, create execution plans, reason through complex problems, use external tools and APIs, access knowledge sources, remember previous interactions, and continuously improve their actions through feedback. These capabilities enable AI systems to complete complex tasks with minimal human intervention.

An Agentic AI system typically combines multiple technologies, including Large Language Models, memory mechanisms, planning algorithms, reasoning techniques, tool integration, Retrieval-Augmented Generation (RAG), vector databases, and workflow orchestration frameworks. Together, these components allow an AI agent to analyze problems, gather information, make informed decisions, and execute tasks efficiently in dynamic environments.

Today, Agentic AI is being adopted across many industries. In healthcare, AI agents assist doctors by analyzing patient records and supporting clinical decision-making. In education, intelligent tutors provide personalized learning experiences. In software development, coding assistants help developers write, debug, and optimize code. Businesses use AI agents for customer support, financial analysis, research automation, document processing, and workflow management. As organizations continue to embrace digital transformation, the demand for professionals with expertise in Agentic AI and LLM technologies is increasing rapidly.

This course, **Agentic AI & LLM Systems**, provides a comprehensive introduction to the concepts, architectures, and practical implementation of modern AI systems. Students will explore the evolution of Natural Language Processing (NLP), understand Transformer models and Large Language Models, learn Prompt Engineering, Retrieval-Augmented Generation (RAG), AI Agents, Model Context Protocol (MCP), and modern Agentic AI frameworks. Through theoretical concepts, practical implementations, industry case studies, and hands-on projects, students will develop the knowledge and skills required to design and build intelligent AI applications capable of solving real-world problems.

Learning Outcomes

After completing this course, students will be able to:

- Explain the fundamentals of Artificial Intelligence, Large Language Models, and Agentic AI.
- Understand the architecture and working principles of Transformer-based language models.
- Differentiate between traditional AI systems, Generative AI, and Agentic AI.
- Design and implement AI-powered applications using modern Agentic AI frameworks.
- Apply Prompt Engineering, Retrieval-Augmented Generation (RAG), and tool integration techniques.
- Develop industry-ready AI solutions using contemporary LLM technologies.

